

Abstract

This study aims to characterize the hydrothermal zone of the summit of Stromboli volcano using ERT, Temperature, Self-Potential and CO₂ flux. First, the summit of the hydrothermal system is mapped using a new 3D inversion code (Fargier, 2011) and data from 13 ERT profiles. Second, the relationship between SP and temperature is investigated using a five years monitoring data set. A linear relationship is observed at different timescales. To our knowledge, such observation has never been emphasized. Finally, this relationship is supported by analogical and numerical experiments of a rising thermal plume in a porous medium.

I – Data profiles

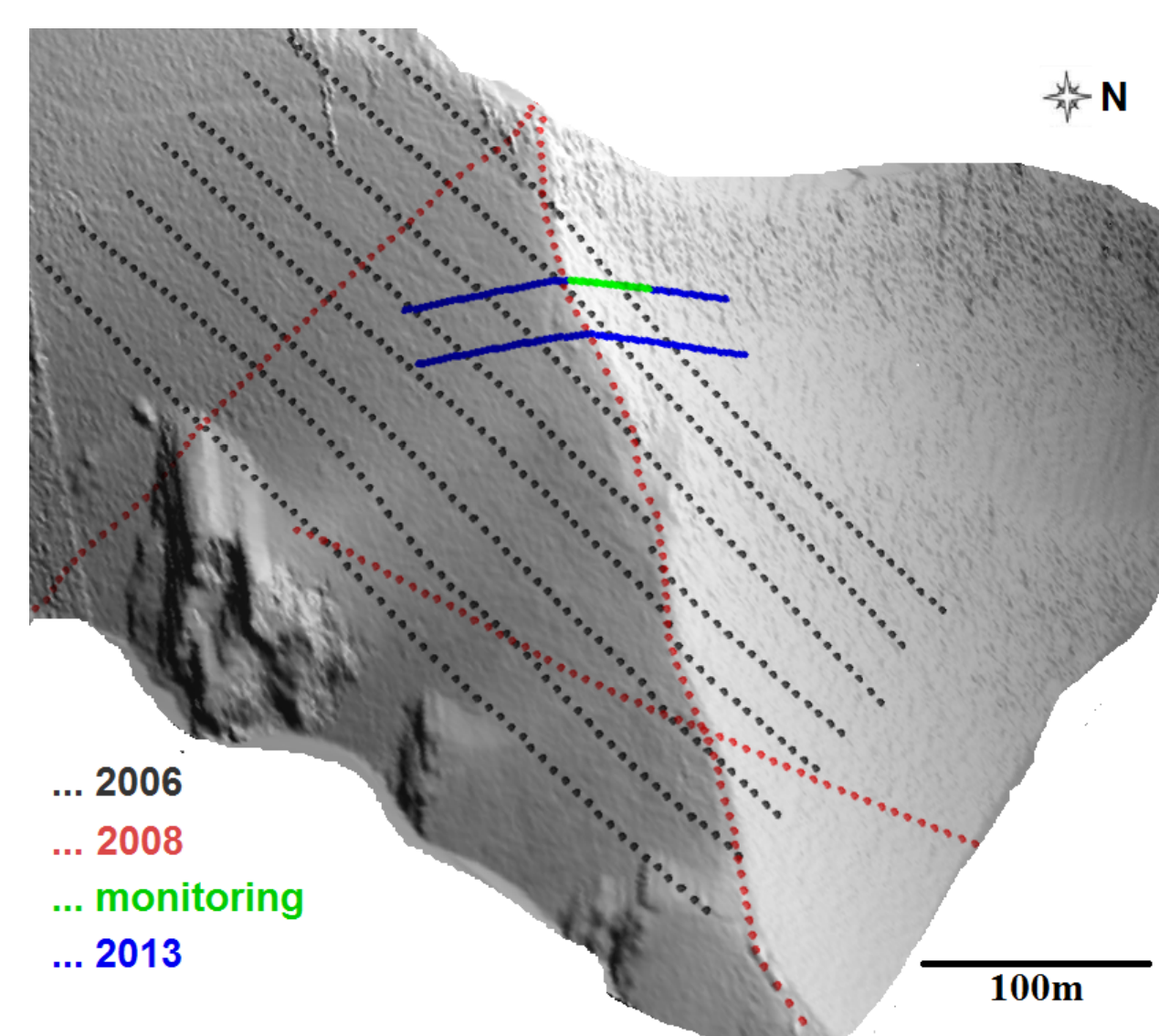


Fig. 1 : Location of SP, Temperature, CO₂ and ERT profiles on a photogrammetric DTM.

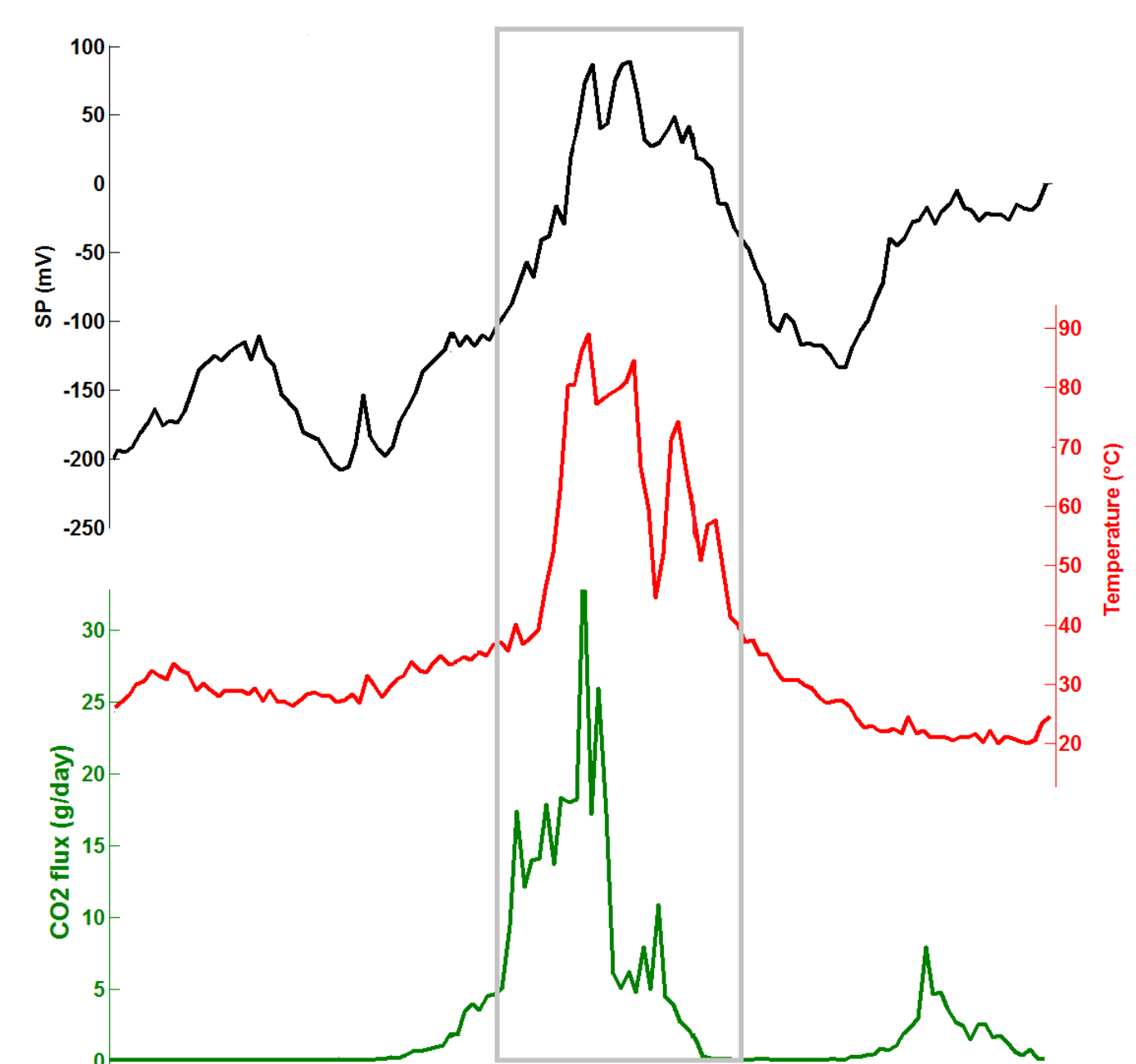


Fig. 2 : SP, Temperature and CO₂ spatial variations along the 2013 western profile.

II- 3D ERT Inversions

Both Res3Dinv (Loke and Barker, 1996) and Colibri43 (Fargier, 2011) inversion models exhibit the same shape for the hydrothermal system.

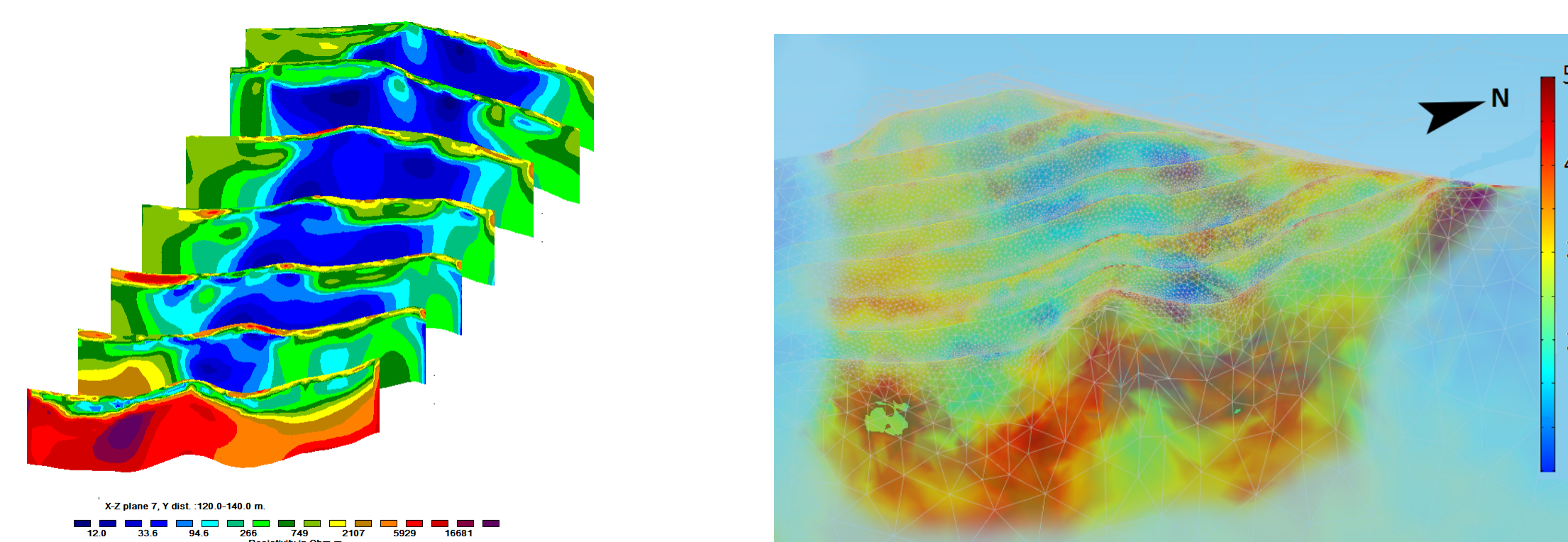


Fig. 3 : a) inversion of 2006 profiles with Res3Dinv (RMS error: 15.4%) and b) Colibri43 (RMS error: 8.17%).

A new selection criterion has been introduced to quantify the accuracy of our model.

DOI : Depth of Investigation (Oldenburg & Li, 1999)

$$DOI(x,y,z) = \frac{m_1^{inv}(x,y,z) - m_2^{inv}(x,y,z)}{DOI_{max}(m_1^{ref}(x,y,z) - m_2^{ref}(x,y,z))}$$

m^{inv} = inverse model - m^{ref} = initial model for two different models (m_1, m_2)

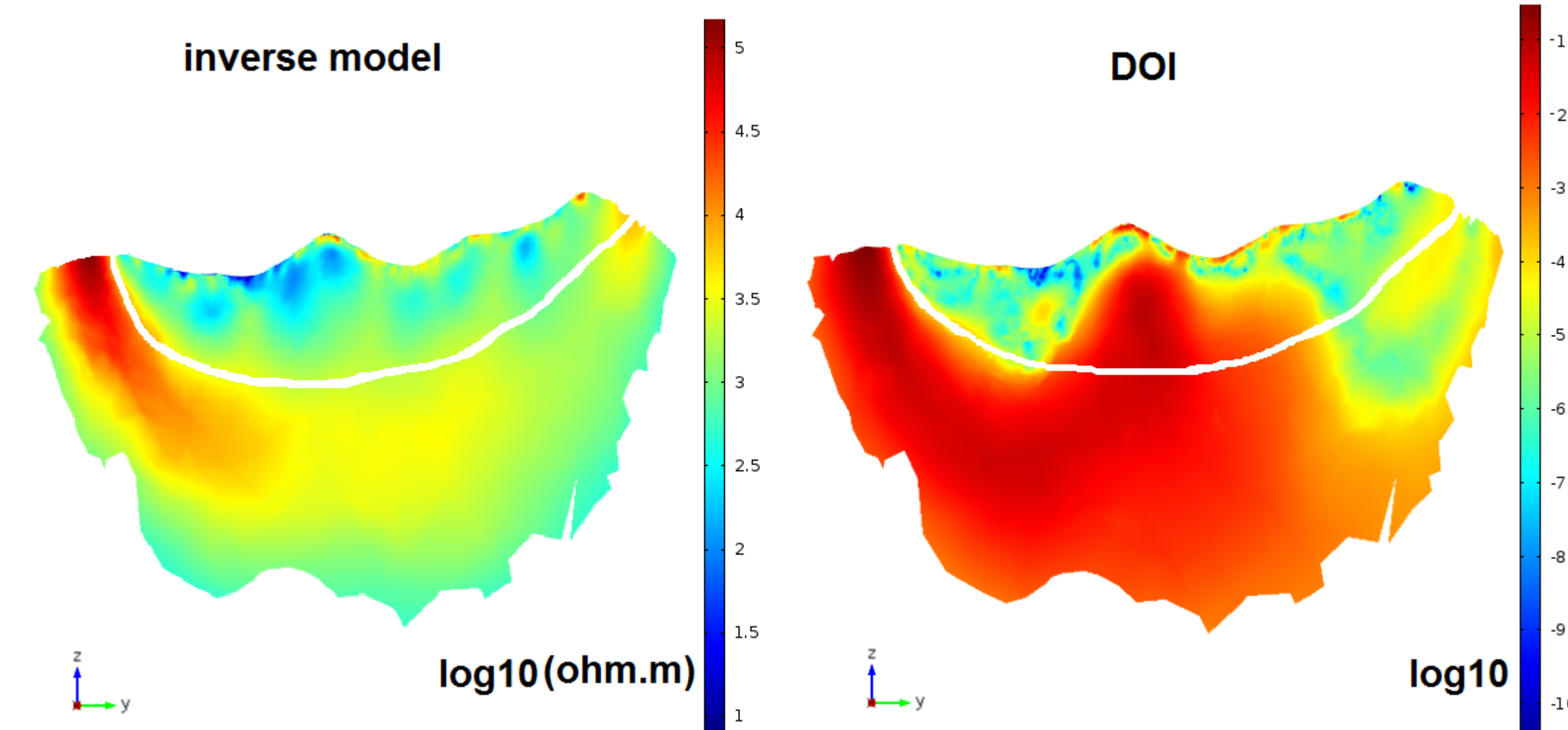


Fig. 4 : a) Inversion model of a profile (2006) showing a high RMS error (>10%) and b) DOI calculated for this profile. The white line refers to the usual DOI (Loke & Barker, 1996).

This suggest that the blue to yellow areas are much more constrained by the measurements that the areas in red.

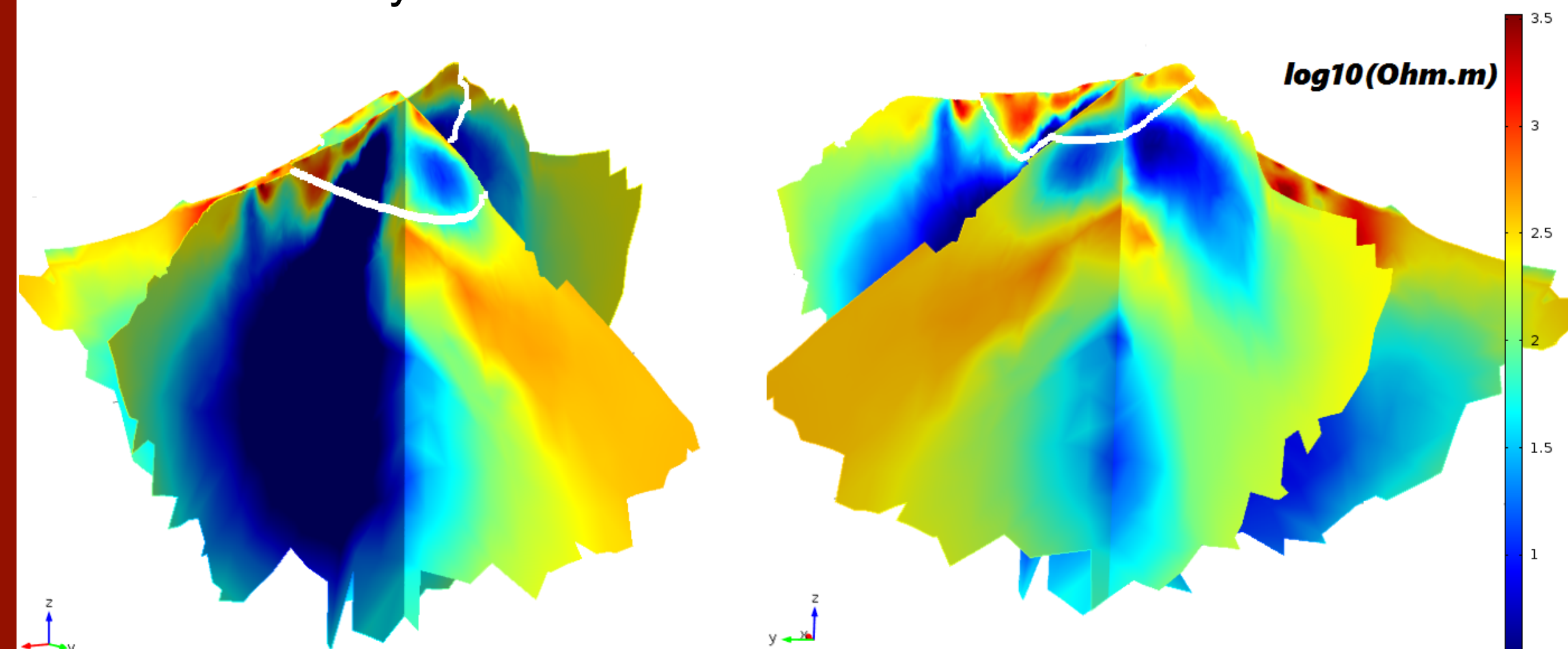


Fig.5 : Preliminary inversion model of 13 profiles performed on 3 different campaigns (2006, 2008, 2013) with Colibri43 (RMS error: 24%). The white line refers to the usual DOI.

III - Relationship between SP and Temperature from monitoring data

Monitoring data reveal a linear relationship between SP and Temperature measurements at different timescale

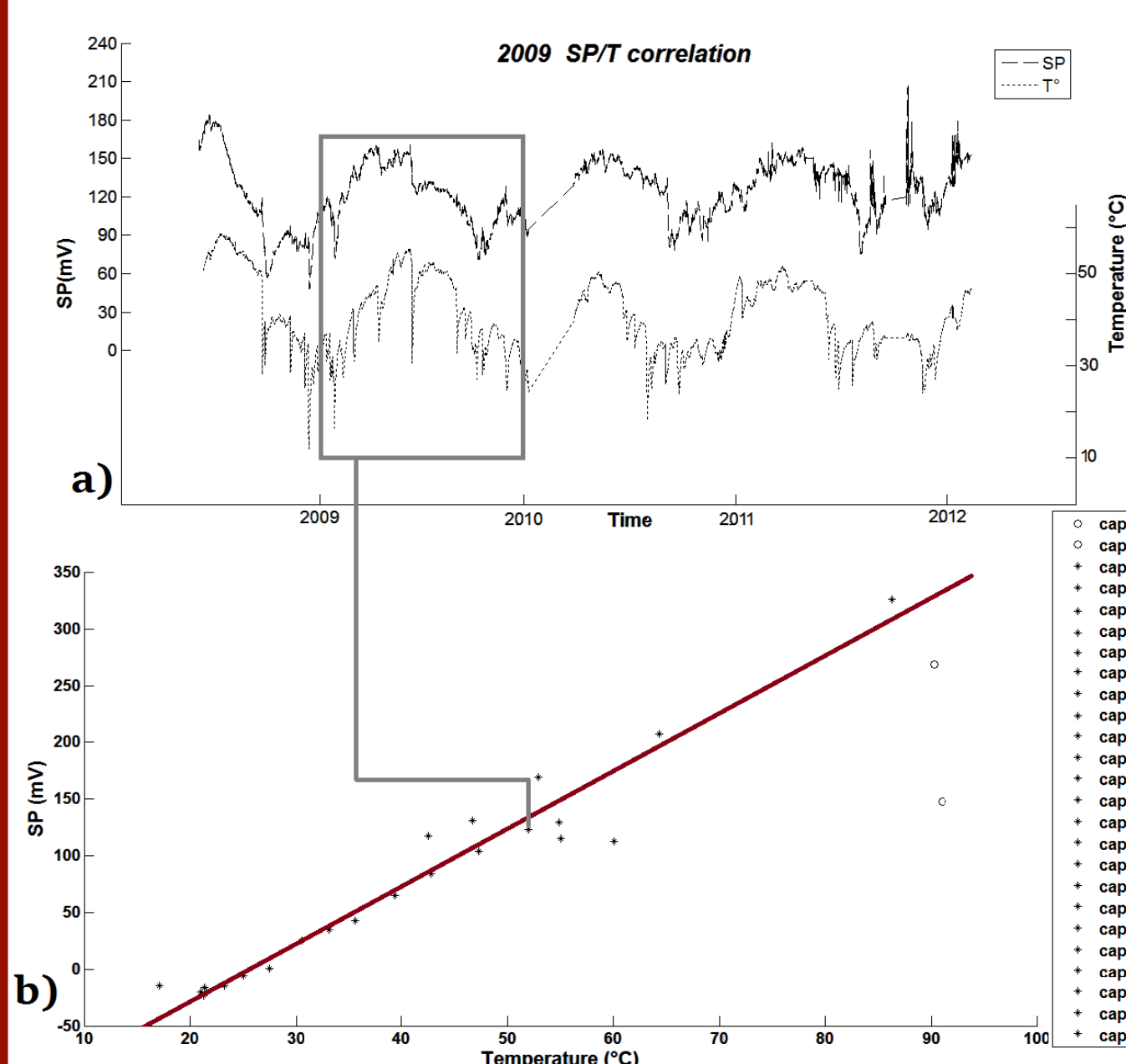


Fig. 6 : a) SP and Temperature evolution for three years (sensor 26) and b) SP mean versus Temperature mean for 2009 and for each sensor.

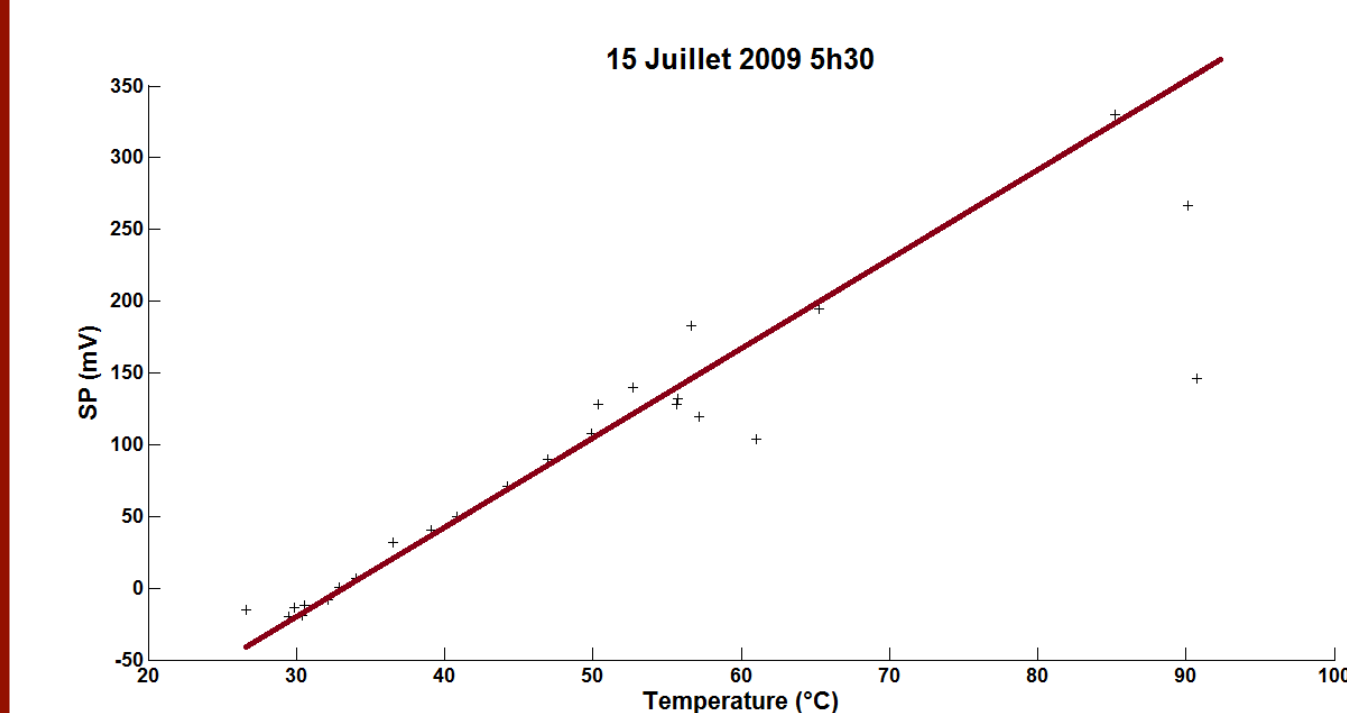


Fig. 7 : SP vs Temperature for each sensor, July, 15th 2009 5h30am. The sensors located outside the curve are always the same.

IV - Laboratory and numerical experiments

Laboratory experiments (R.Antoine, 2011) show the same relationship between SP and Temperature.

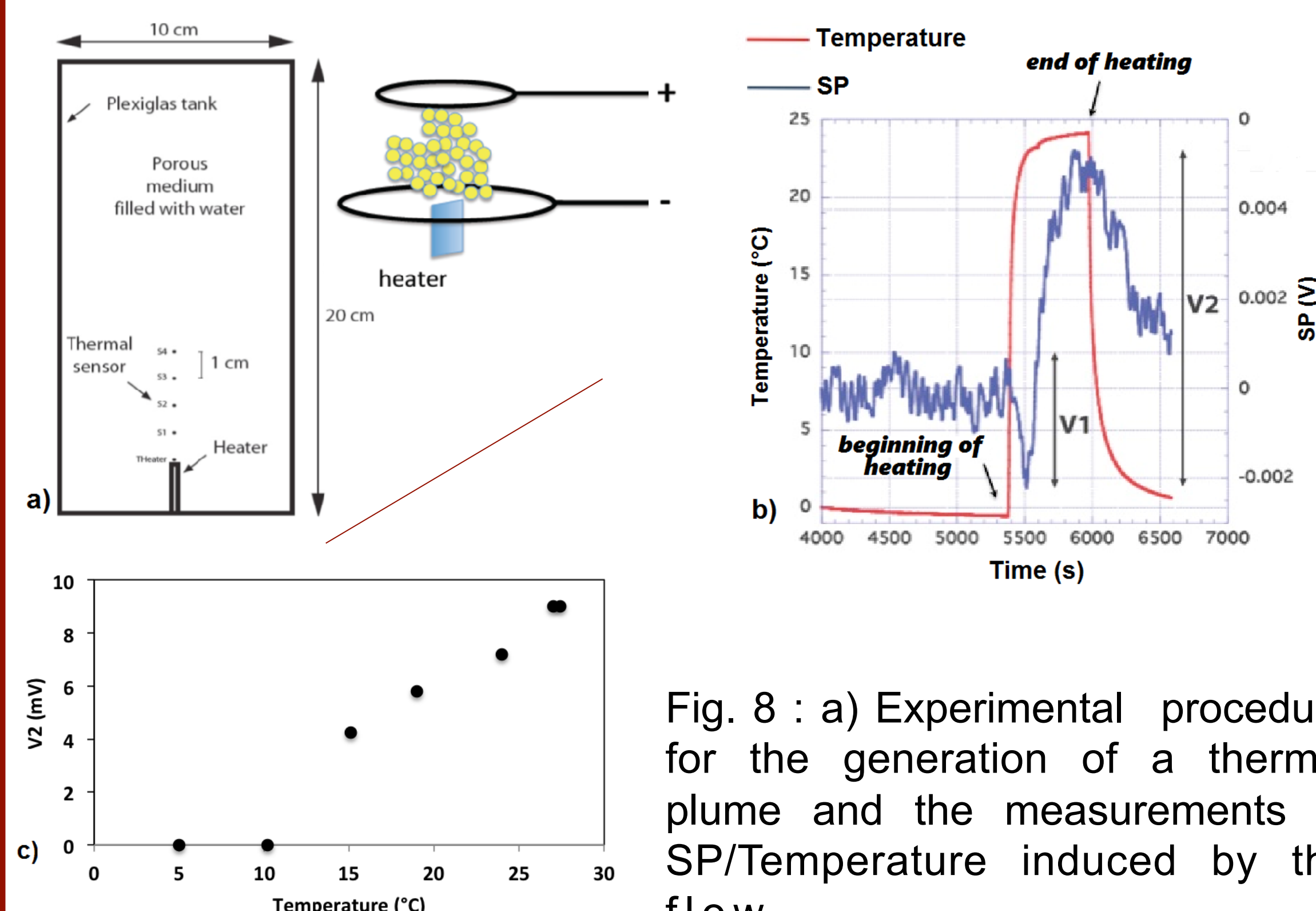


Fig. 8 : a) Experimental procedure for the generation of a thermal plume and the measurements of SP/Temperature induced by the flow. b) Temperature and SP as a function of time. c) Temperature vs SP for a power of 4.49W. V1 is the same for different applied powers. V2 linearly increases with the power and the temperature.

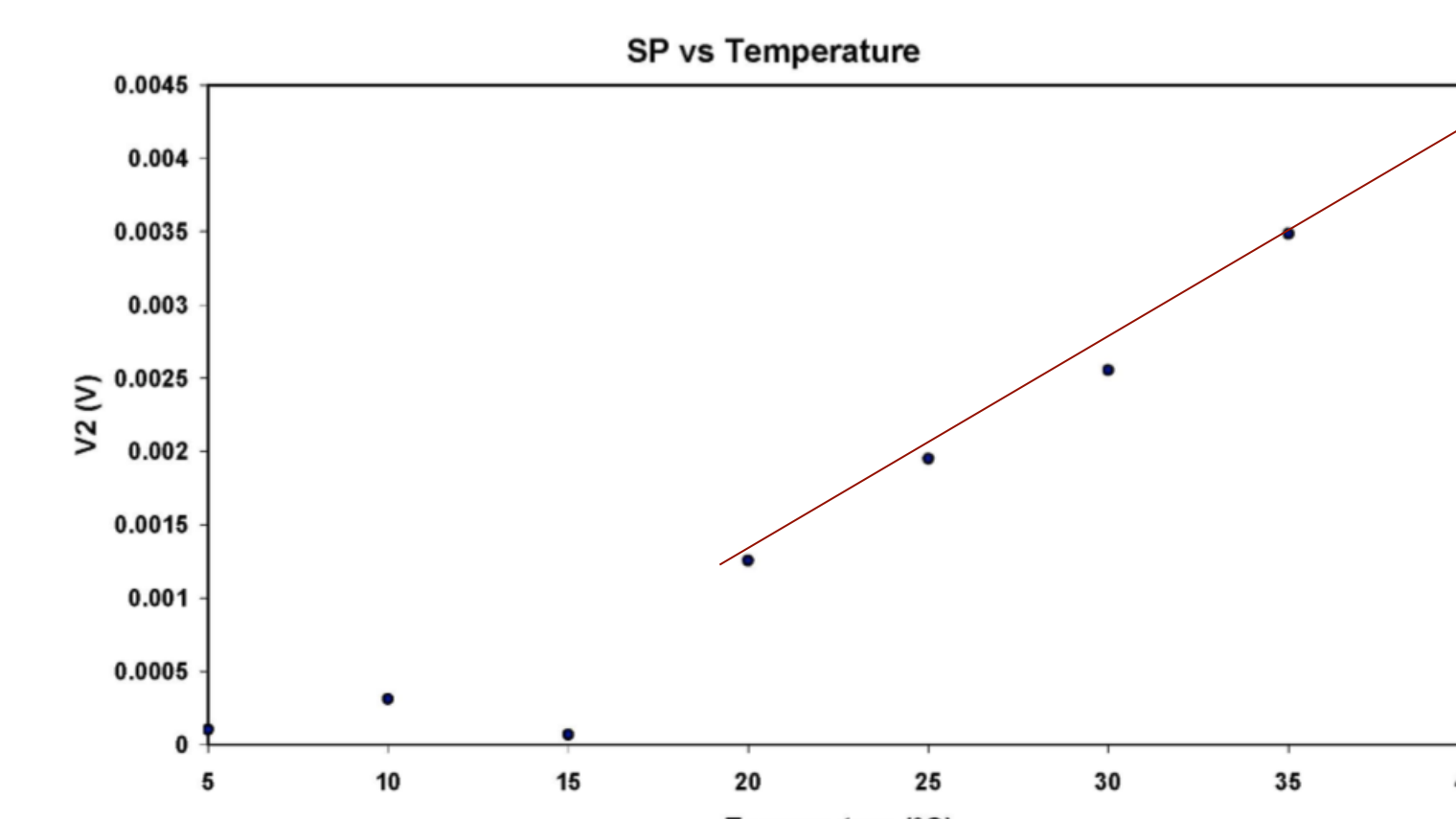


Fig. 9 : Numerical simulation of a thermal plume in a porous medium.

Conclusions

- For the first time on a volcano, we present a simultaneous inversion of 13 ERT profiles. This allows us mapping the summit of the hydrothermal system of Stromboli. The CO₂, T and SP mapping corroborate spatially the ERT inversions and locate the summit of the hydrothermal system of Stromboli.
- A linear relationship between SP and Temperature is emphasized from monitoring data and laboratory/numerical experiments.

Perspectives

- Evaluation of the accuracy of the simultaneous inversion compared to a cascade inversion.
- Characterization of the link between Darcy velocity in porous media and SP/T relationship ; estimation of the influence of the permeability and the non boussinesq effects on this relationship.
- Establishment of a new protocol for geothermal exploration.

Acknowledgments : This work is supported by EDF

References :

- Y.Fargier, 2011, Ph.D. at IFSTTAR (Nantes, France).
Loke & Barker, 1996 Geophysical Prospecting, 44(1), 131-152.
Oldenburg & Li, 1999, Geophysics 64(2), 403-416.
Antoine R. Kurita K. - 2011 Geofluids